Linear classifiers: Parameter learning

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Learn a probabilistic classification model

"The sushi & everything else were awesome!"

Definite +1

$P(y=+1|\mathbf{x}=$ *"The sushi & everything else were awesome!"* $) = 0.99$

"The sushi was good, the service was OK"

Not sure

$P(y=+1|\mathbf{x}=$ *"The sushi was good, the service was OK"* $) = 0.55$

Many classifiers provide a degree of certainty:

Output label

Input sentence

$$P(y|\mathbf{x})$$

Extremely useful in practice

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A (linear) classifier

- Will use training data to learn a weight or coefficient for each word

Word	Coefficient	Value
	$\hat{w}_0$	-2.0
good	$\hat{w}_1$	1.0
great	$\hat{w}_2$	1.5
awesome	$\hat{w}_3$	2.7
bad	$\hat{w}_4$	-1.0
terrible	$\hat{w}_5$	-2.1
awful	$\hat{w}_6$	-3.3
restaurant, the, we, …	$\hat{w}_7, \hat{w}_8, \hat{w}_9, \ldots$	0.0
…		…

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Logistic regression model

$$\text{Score}(\mathbf{x}_i) = \hat{\mathbf{w}}^T h(\mathbf{x}_i)$$

$-\infty$ ⟷ $+\infty$

0.0

0.0 ⟷ 0.5 ⟷ 1.0

$$\hat{P}(y=+1|\mathbf{x},\hat{\mathbf{w}}) = \frac{1}{1 + e^{-\hat{\mathbf{w}}^T h(\mathbf{x})}}$$

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Quality metric for logistic regression: Maximum likelihood estimation

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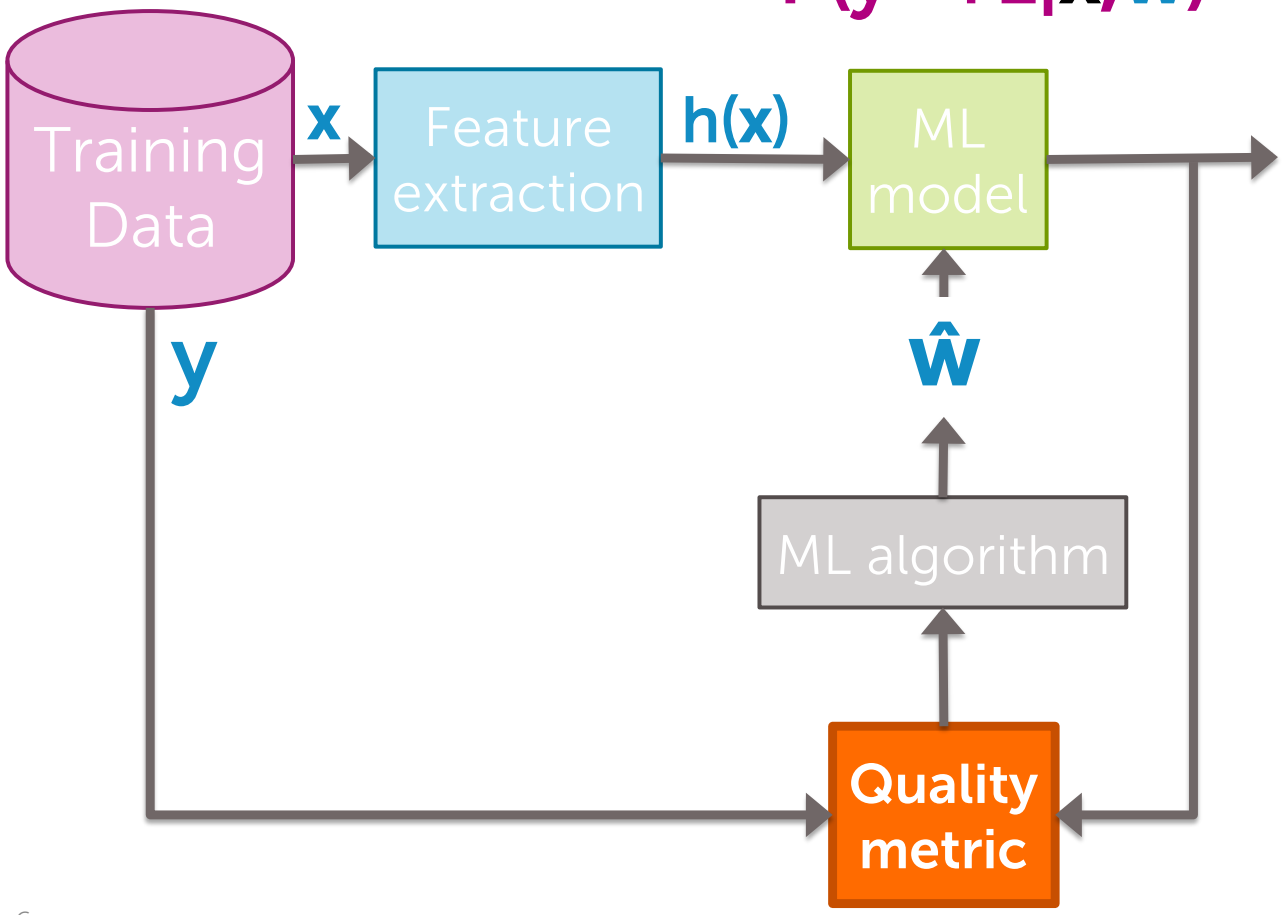
$$\hat{P}(y=+1|\mathbf{x},\hat{\mathbf{w}}) = \frac{1}{1+e^{-\hat{\mathbf{w}}^T h(\mathbf{x})}}$$



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Learning problem

Training data:
N observations $(\mathbf{x}_i, y_i)$

x[1] = #awesome	x[2] = #awful	y = sentiment
2	1	+1
0	2	-1
3	3	-1
4	1	+1
1	1	+1
2	4	-1
0	3	-1
0	1	-1
2	1	+1

Optimize **quality metric** on training data

$\hat{\mathbf{w}}$

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Finding best coefficients

x[1] = #awesome	**x**[2] = #awful	y = sentiment
2	1	+1
0	2	-1
3	3	-1
4	1	+1
1	1	+1
2	4	-1
0	3	-1
0	1	-1
2	1	+1

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Finding best coefficients

x[1] = #awesome	**x**[2] = #awful	y = sentiment
0	2	-1
3	3	-1
2	4	-1
0	3	-1
0	1	-1
2	4	-1
0	3	-1
0	1	-1

x[1] = #awesome	**x**[2] = #awful	y = sentiment
2	1	+1
4	1	+1
1	1	+1
2	1	+1
1	1	+1
2	1	+1

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Finding best coefficients

x[1] = #awesome	x[2] = #awful	y = sentiment
0	2	-1
3	3	-1
2	4	-1
0	3	-1
0	1	-1

$P(y=+1|\mathbf{x}_i,\mathbf{w}) = 0.0$

x[1] = #awesome	x[2] = #awful	y = sentiment
2	1	+1
4	1	+1
1	1	+1
2	1	+1

$P(y=+1|\mathbf{x}_i,\mathbf{w}) = 1.0$

Pick $\hat{\mathbf{w}}$ that makes

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Quality metric = Likelihood function

Negative data points

Positive data points

$P(y=+1|\mathbf{x}_i,\mathbf{w}) = 0.0$

$P(y=+1|\mathbf{x}_i,\mathbf{w}) = 1.0$

No $\hat{\mathbf{w}}$ achieves perfect predictions (usually)

Likelihood $\ell(\mathbf{w})$: Measures quality of fit for model with coefficients $\mathbf{w}$

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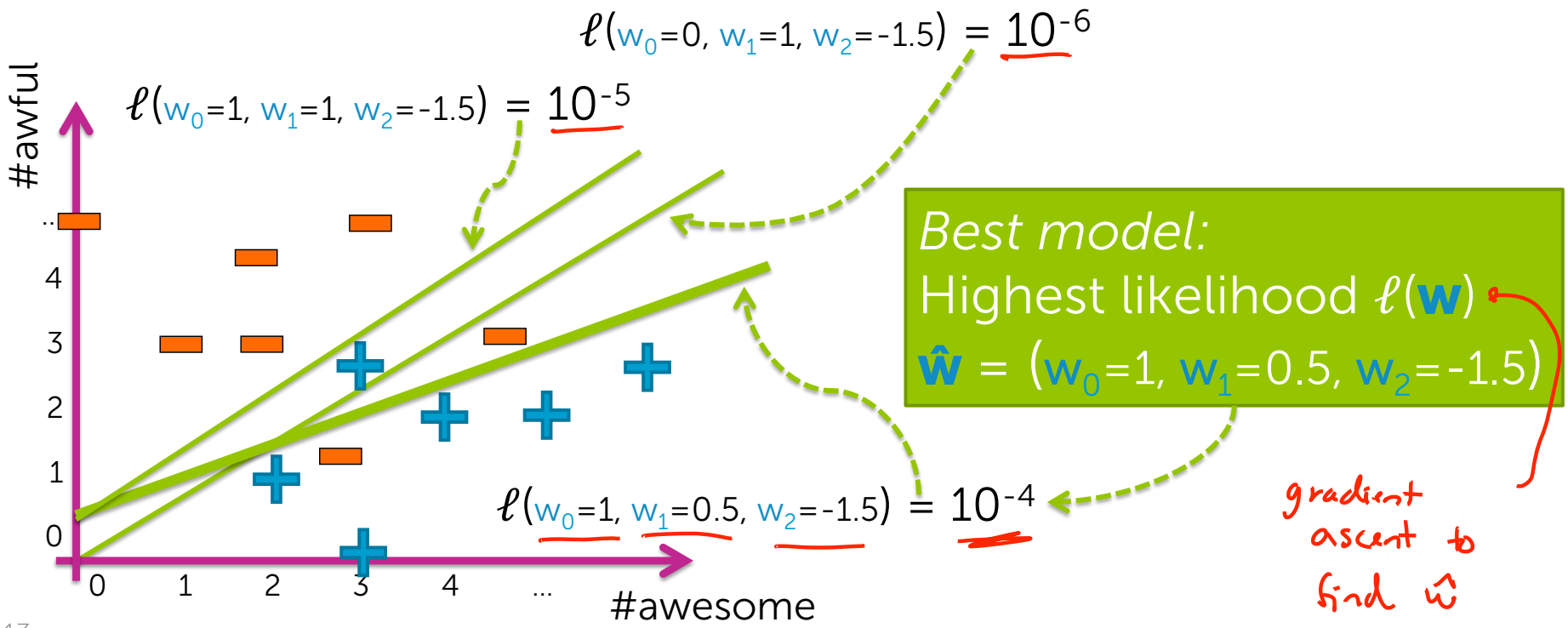
Find "best" classifier

Maximize likelihood over all possible w_0, w_1, w_2



Best model:
Highest likelihood $\ell(\mathbf{w})$
$\hat{\mathbf{w}} = (w_0=1, w_1=0.5, w_2=-1.5)$

gradient ascent to find $\hat{w}$

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Data likelihood

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Quality metric: probability of data

x[1] = #awesome	x[2] = #awful	y = sentiment
2	1	+1

x_1 $y_1 =$

If model good, should predict:

$\hat{y}_1 = +1$

Pick **w** to maximize:

$P(y=+1 \mid x_1, w) = P(y=+1 \mid x[1]=2, x[2]=1, w)$

x[1] = #awesome	x[2] = #awful	y = sentiment
0	2	-1

$x_2 =$ $y_2 =$

If model good, should predict:

$\hat{y}_2 = -1$

Pick **w** to maximize:

$P(y=-1 \mid x_2, w)$

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Maximizing likelihood (probability of data)

Data point	x[1]	x[2]	y	Choose w to maximize
$\mathbf{x}_1, y_1$	2	1	+1	$P(y=+1 \mid x_1, w) = P(y=+1 \mid x[1]=2, x[2]=1, w)$
$\mathbf{x}_2, y_2$	0	2	-1	$P(y=-1 \mid x_2, w)$
$\mathbf{x}_3, y_3$	3	3	-1	$P(y=-1 \mid x_3, w)$
$\mathbf{x}_4, y_4$	4	1	+1	$P(y=+1 \mid x_4, w)$
$\mathbf{x}_5, y_5$	1	1	+1	
$\mathbf{x}_6, y_6$	2	4	-1	
$\mathbf{x}_7, y_7$	0	3	-1	
$\mathbf{x}_8, y_8$	0	1	-1	
$\mathbf{x}_9, y_9$	2	1	+1	

Must combine into single measure of quality ?

Multiply probabilities

$P(y=+1 \mid x_1, w)\, P(y=-1 \mid x_2, w)\, P(y=-1 \mid x_3, w) \ldots$

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Learn logistic regression model with maximum likelihood estimation (MLE)

Data point	x[1]	x[2]	y	Choose **w** to maximize
$\mathbf{x}_1, y_1$	2	1	y_1: +1	$P(y=+1 \mid \mathbf{x}[1]=2, \mathbf{x}[2]=1, \mathbf{w})$
$\mathbf{x}_2, y_2$	0	2	-1	$P(y=-1 \mid \mathbf{x}[1]=0, \mathbf{x}[2]=2, \mathbf{w})$
$\mathbf{x}_3, y_3$	3	3	-1	$P(y=-1 \mid \mathbf{x}[1]=3, \mathbf{x}[2]=3, \mathbf{w})$
$\mathbf{x}_4, y_4$	4	1	+1	$P(y=+1 \mid \mathbf{x}[1]=4, \mathbf{x}[2]=1, \mathbf{w})$

$$\ell(\mathbf{w}) = \underbrace{P(y_1 \mid \mathbf{x}_1, \mathbf{w}) \quad P(y_2 \mid \mathbf{x}_2, \mathbf{w}) \quad P(y_3 \mid \mathbf{x}_3, \mathbf{w}) \quad P(y_4 \mid \mathbf{x}_4, \mathbf{w})}$$

num. of data points

$$\ell(w) = \prod_{i=1}^{N} P(y_i \mid \mathbf{x}_i, \mathbf{w})$$

← pick w to make this fn. as large as possible

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Finding best linear classifier with gradient ascent

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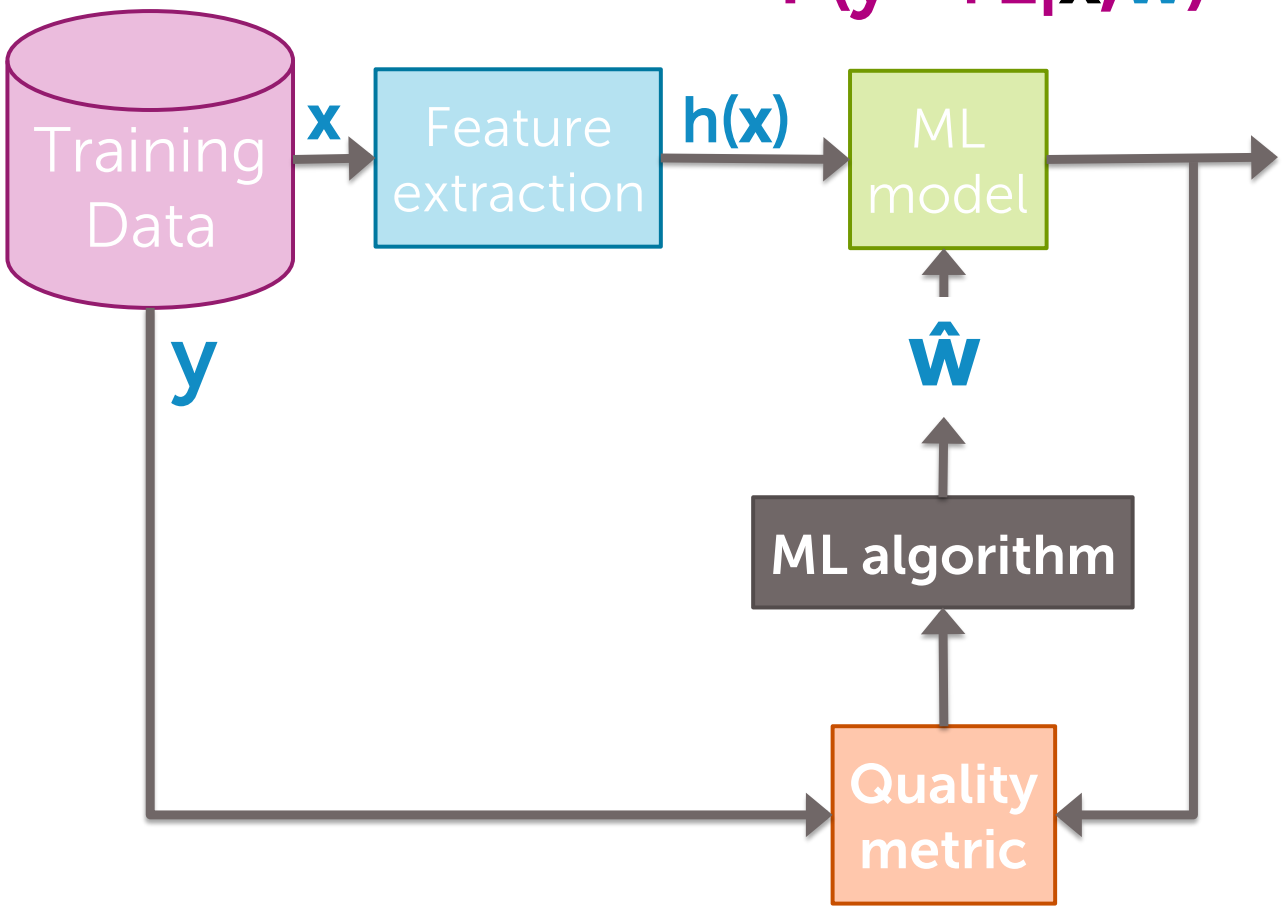
$$\hat{P}(y=+1|\mathbf{x},\hat{\mathbf{w}}) = \frac{1}{1+e^{-\hat{\mathbf{w}}^T h(\mathbf{x})}}$$



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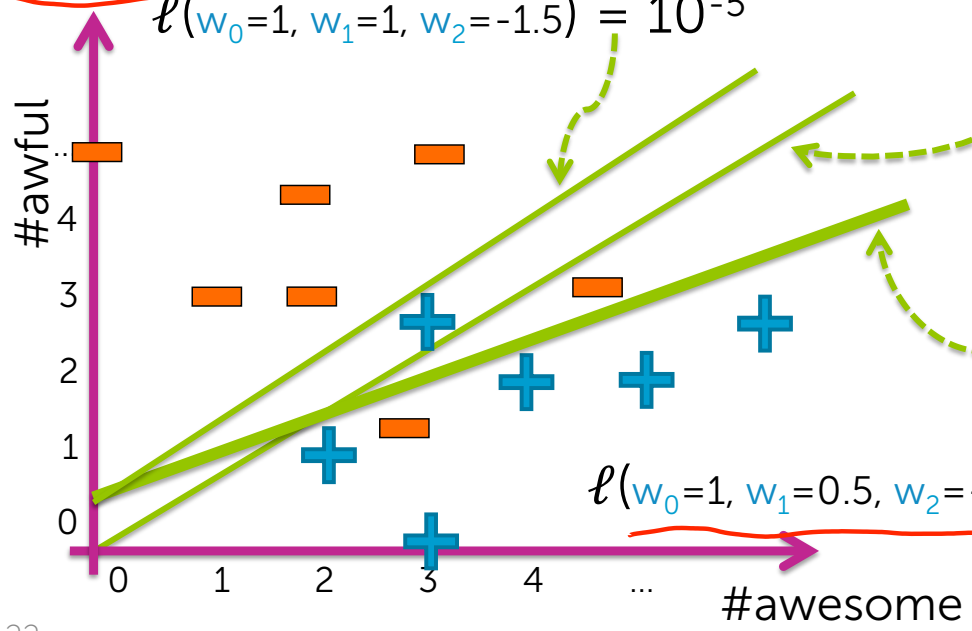
Find "best" classifier

Maximize likelihood over all possible w_0, w_1, w_2

$$\ell(\mathbf{w}) = \prod_{i=1}^{N} P(y_i \mid \mathbf{x}_i, \mathbf{w})$$

$\ell(w_0=0, w_1=1, w_2=-1.5) = 10^{-6}$

$\ell(w_0=1, w_1=1, w_2=-1.5) = 10^{-5}$



$\ell(w_0=1, w_1=0.5, w_2=-1.5) = 10^{-4}$

Best model:
Highest likelihood $\ell(\mathbf{w})$
$\hat{\mathbf{w}} = (w_0=1, w_1=0.5, w_2=-1.5)$

optimize with gradient ascent

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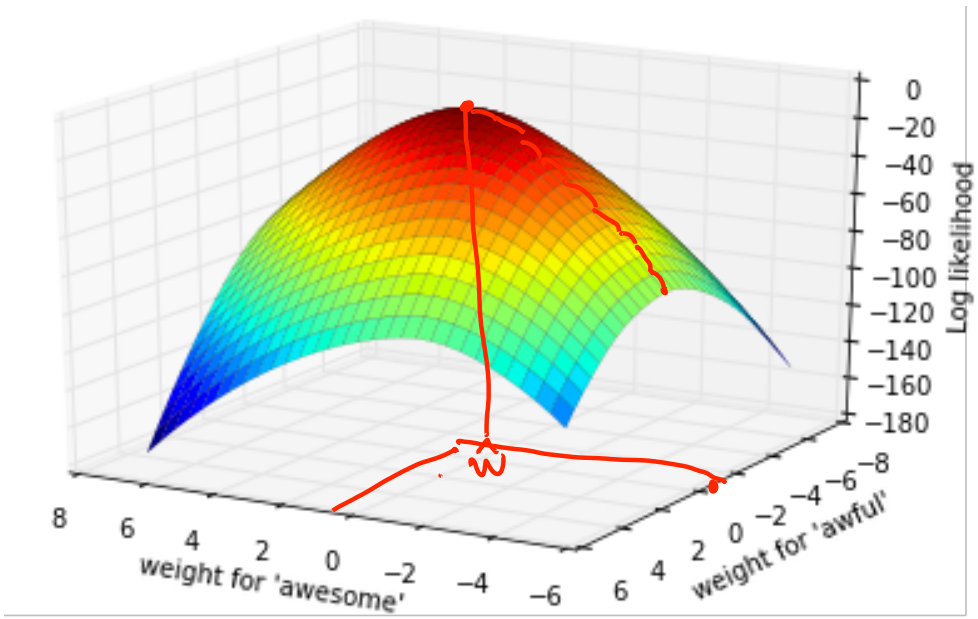
Maximizing likelihood



Maximize function over all possible w_0, w_1, w_2

$$\max_{w_0,w_1,w_2} \underbrace{\prod_{i=1}^{N} P(y_i \mid \mathbf{x}_i, \mathbf{w})}$$

$\ell(w_0, w_1, w_2)$ is a function of 3 variables

No closed-form solution ➔ use gradient ascent

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Review of gradient ascent

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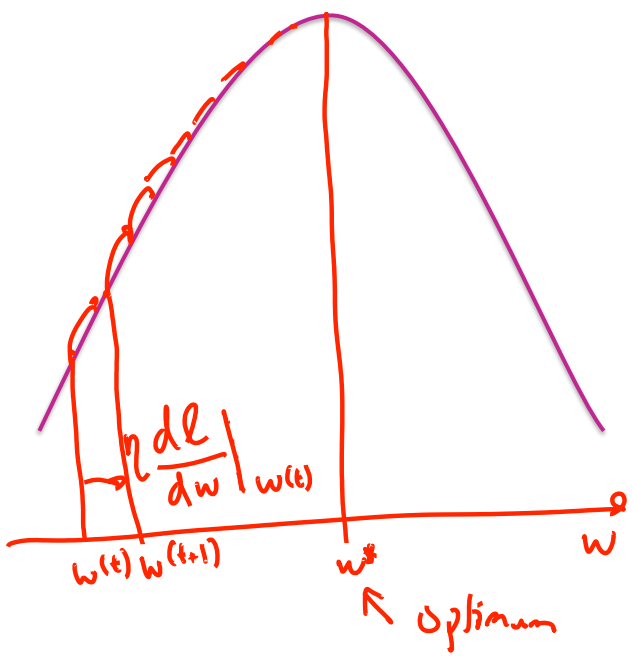
Finding the max via hill climbing



Algorithm:

while not converged

$$w^{(t+1)} \leftarrow w^{(t)} + \eta \frac{d\ell}{dw}\Big|_{w^{(t)}}$$

step size

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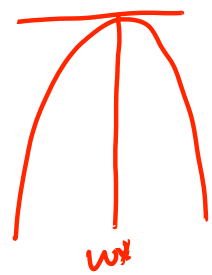
Convergence criteria

For convex functions, optimum occurs when

$$\frac{d\ell}{dw} = 0$$



In practice, stop when

$$\left.\frac{d\ell}{dw}\right|_{w^{(t)}} < \varepsilon$$

tolerance

Algorithm:

while not converged

$$w^{(t+1)} \leftarrow w^{(t)} + \eta \left.\frac{d\ell}{dw}\right|_{w^{(t)}}$$

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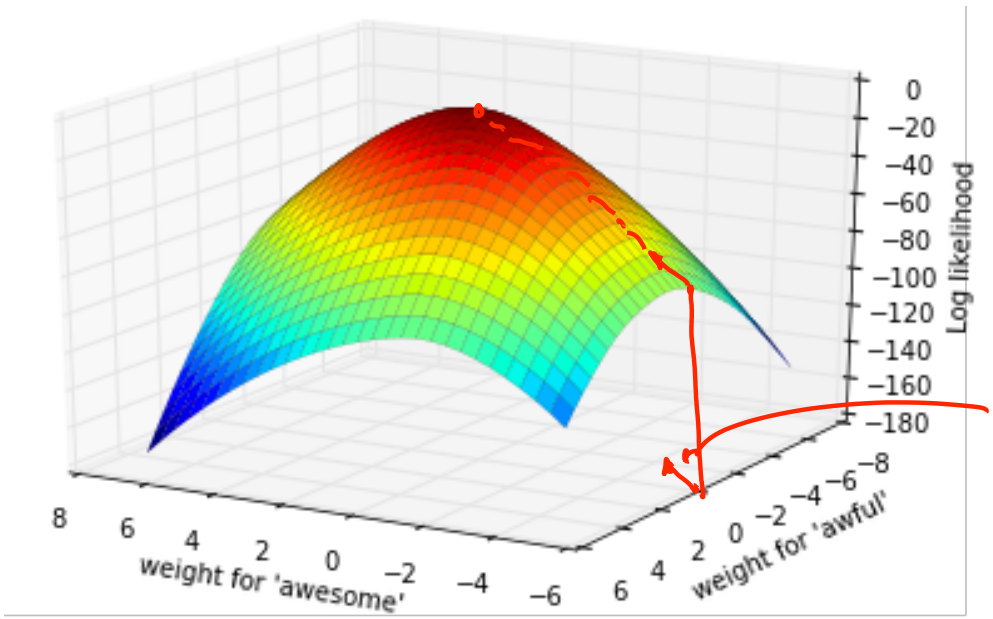
Moving to multiple dimensions: Gradients



$$\nabla \ell(\mathbf{w}) = \begin{bmatrix} \frac{\partial \ell}{\partial w_0} \\ \frac{\partial \ell}{\partial w_1} \\ \vdots \\ \frac{\partial \ell}{\partial w_D} \end{bmatrix}$$

D+1 dim vector

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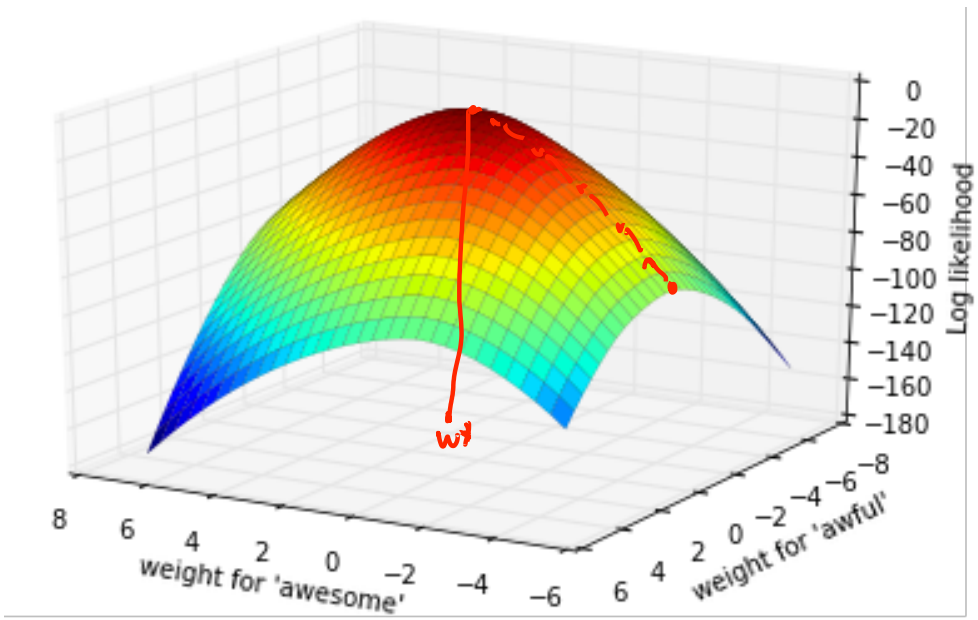
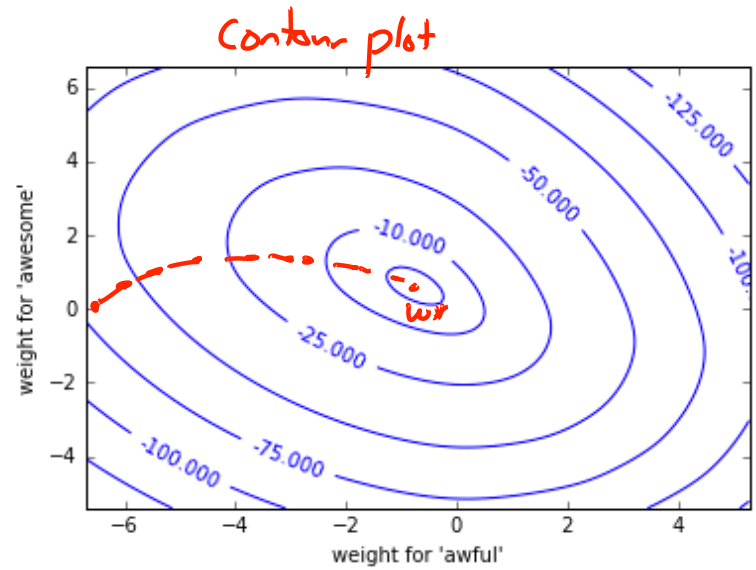
Contour plots





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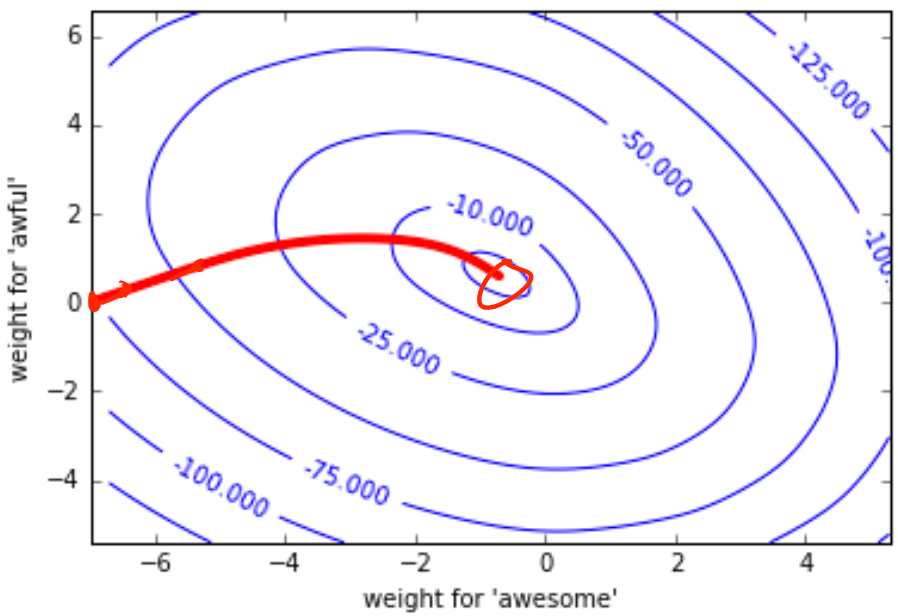
Gradient ascent



Algorithm:

$w^{(0)} = 0$, random, or something smart.

while not converged

$$\mathbf{w}^{(t+1)} \leftarrow \mathbf{w}^{(t)} + \eta \nabla \ell(\mathbf{w}^{(t)})$$

step size

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Learning algorithm for logistic regression

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Derivative of (log-)likelihood

Sum over data points

Feature value

Difference between truth and prediction

$$\frac{\partial \ell(\mathbf{w})}{\partial \mathbf{w}_j} = \sum_{i=1}^{N} h_j(\mathbf{x}_i)\Big(\mathbb{1}[y_i = +1] - \underbrace{P(y = +1 \mid \mathbf{x}_i, \mathbf{w})}_{\text{predict } x_i \text{ is positive}}\Big)$$

Indicator function:

$$\mathbb{1}[y_i = +1] = \begin{cases} 1 & \text{if } y_i = +1 \\ 0 & \text{if } y_i = -1 \end{cases}$$

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Computing derivative

$$\frac{\partial \ell(\mathbf{w}^{(t)})}{\partial \mathbf{w}_j} = \sum_{i=1}^{N} h_j(\mathbf{x}_i)\Big(\mathbb{1}[y_i = +1] - P(y = +1 \mid \mathbf{x}_i, \mathbf{w}^{(t)})\Big)$$

$w^{(t)}$:

$w_0^{(t)}$	0
$w_1^{(t)}$	1
$w_2^{(t)}$	-2

$\frac{\partial \ell}{\partial w_1}$

$h_1(x)$ = # awesome

x[1]	x[2]	y	P(y=+1|x_i,w)	Contribution to derivative for w_1
2	1	+1	0.5	$2(1-0.5) = 1$
0	2	-1	0.02	$0(0-0.02) = 0$
3	3	-1	0.05	$3(0-0.05) = -0.15$
4	1	+1	0.88	$4(1-0.88) = 0.48$

Total derivative:

$$\frac{\partial \ell(w^{(t)})}{\partial w_1} = 1 + 0 - 0.15 + 0.48 = 1.33$$

$$w_1^{(t+1)} = w_1^{(t)} + \eta \frac{\partial \ell(w^{(t)})}{\partial w_1} \quad \Big| \quad \eta = 0.1$$

$$= 1 + 0.1 \times 1.33 = 1.133$$

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Derivative of (log-)likelihood: Interpretation

Sum over data points

Feature value

Difference between truth and prediction

$$\frac{\partial \ell(\mathbf{w})}{\partial \mathbf{w}_j} = \sum_{i=1}^{N} \underbrace{h_j(\mathbf{x}_i)\Big(\mathbb{1}[y_i = +1] - P(y = +1 \mid \mathbf{x}_i, \mathbf{w})\Big)}_{\Delta_i}$$

If $h_j(\mathbf{x}_i)=1$:	$P(y=+1\|x_i,w) \approx 1$	$P(y=+1\|x_i,w) \approx 0$
$y_i=+1$	$\Delta_i = (1-1) \approx 0$ ↳ don't change anything!	$\Delta_i \approx 1 \Rightarrow$ increase w_j $\Rightarrow$ Score(x_i) becomes larger $\Rightarrow$ $P(y=+1\|x_i,w)$ increases
$y_i=-1$	$\Delta_i = -1 \Rightarrow w_j$ to decrease $\Rightarrow$ Score(x_i) decreases $\Rightarrow$ $P(y=+1\|x_i,w)$ decrease	$\Delta_i \approx 0$ $\Rightarrow$ don't change anything

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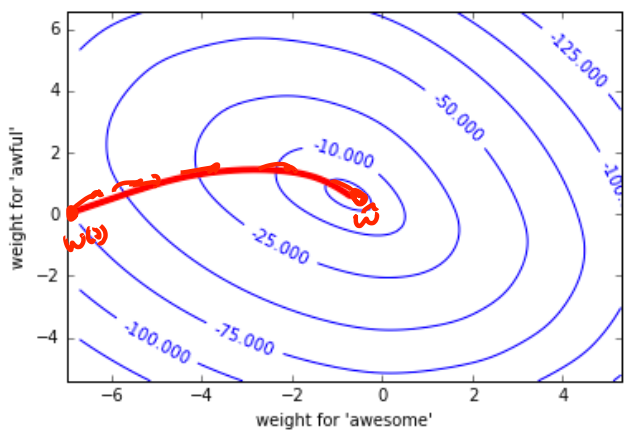
Summary of gradient ascent for logistic regression



init $\mathbf{w}^{(1)}=0$ (or randomly, or smartly), $t=1$

while $||\nabla \ell(\mathbf{w}^{(t)})|| > \varepsilon$ ← tolerance

$$\frac{1}{1+e^{-\mathbf{w}^{(t)T}h(x_i)}}$$

for $j=0,\ldots,D$ ← coefficient

partial[j] = $\sum_{i=1}^{N} h_j(\mathbf{x}_i)\Big(\mathbb{1}[y_i=+1] - P(y=+1 \mid \mathbf{x}_i, \mathbf{w}^{(t)})\Big)$

$w_j^{(t+1)} \leftarrow w_j^{(t)} + \eta$ partial[j]

$t \leftarrow t+1$

step size

$$\frac{\partial \ell(w^{(t)})}{\partial w_j}$$

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Choosing the step size η

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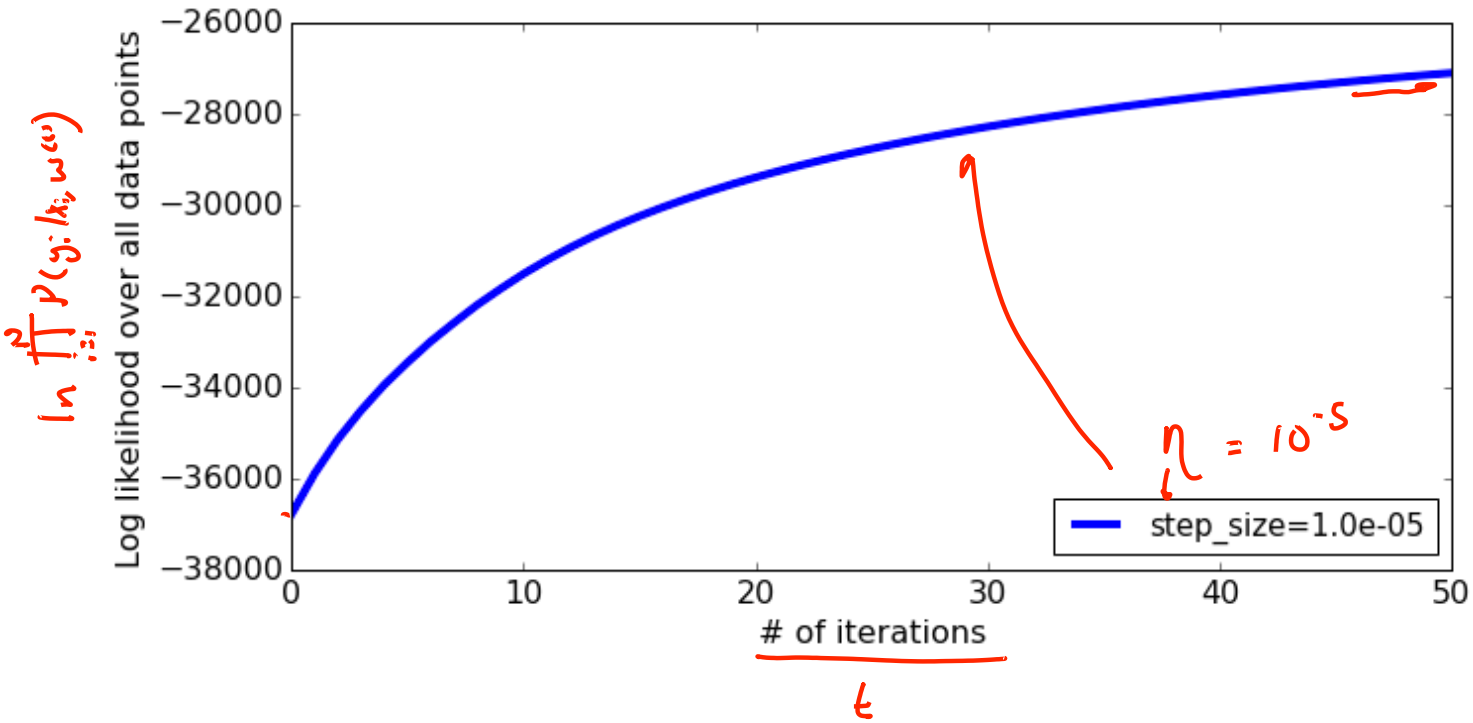
Learning curve:
Plot quality (likelihood) over iterations



need more than 50 iterations

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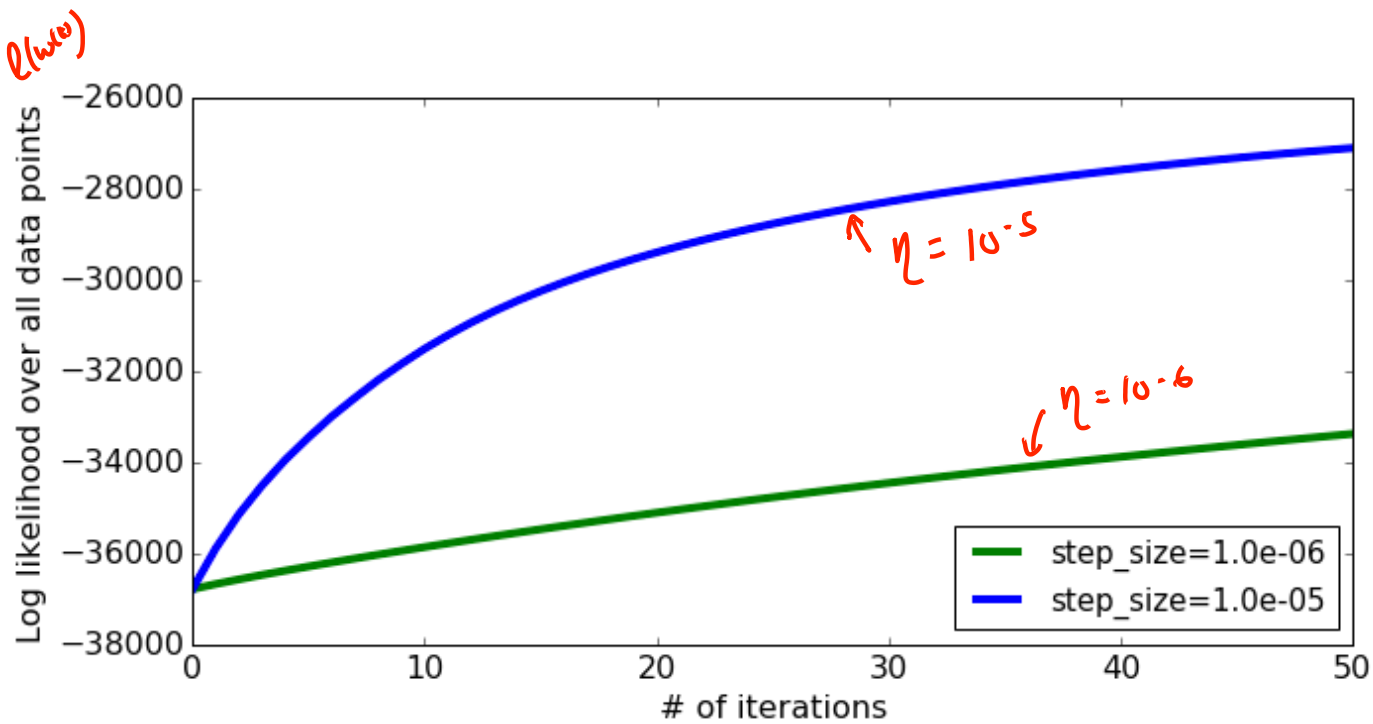
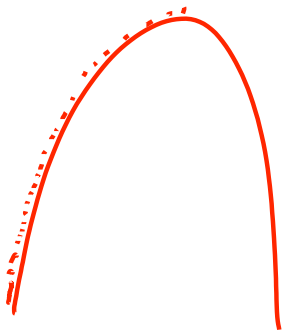
If step size is too small, can take a long time to converge





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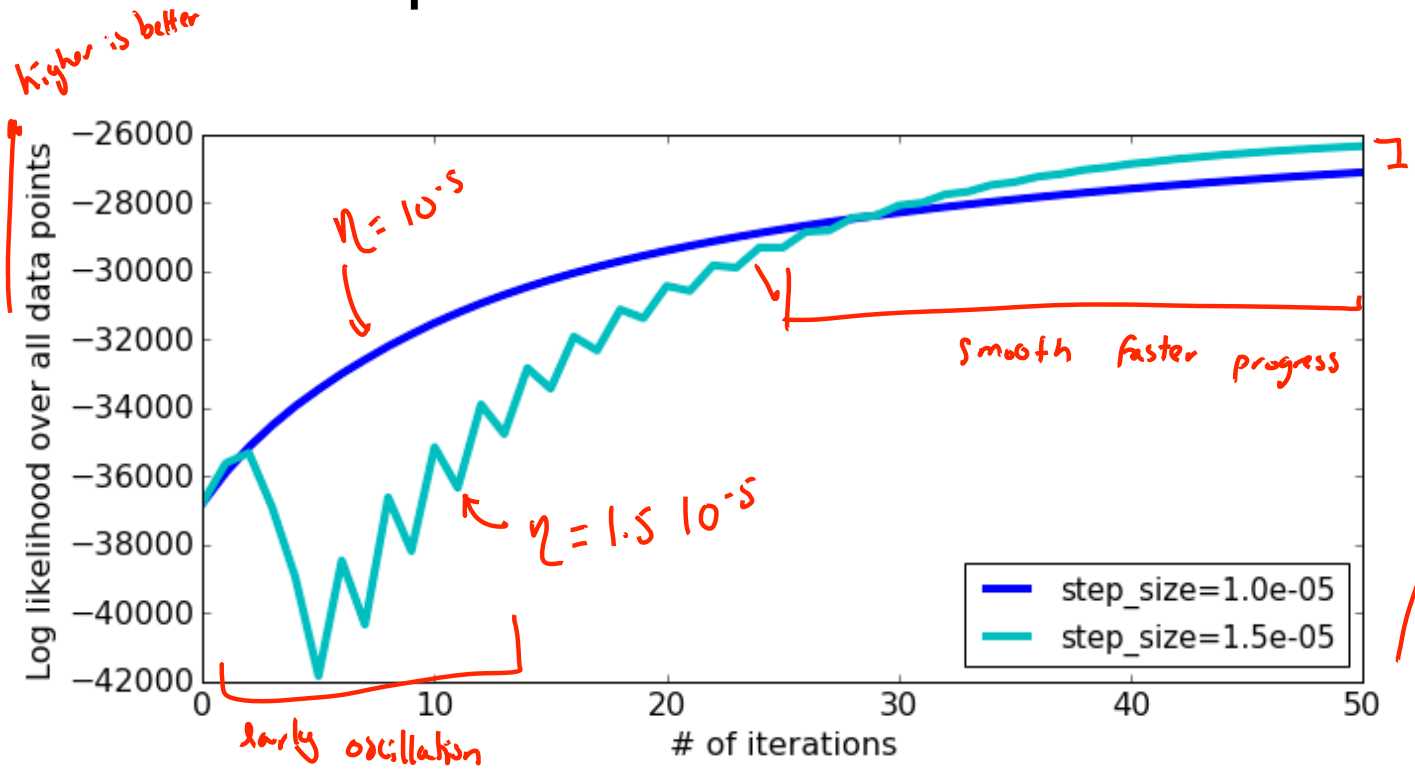
Compare converge with different step sizes



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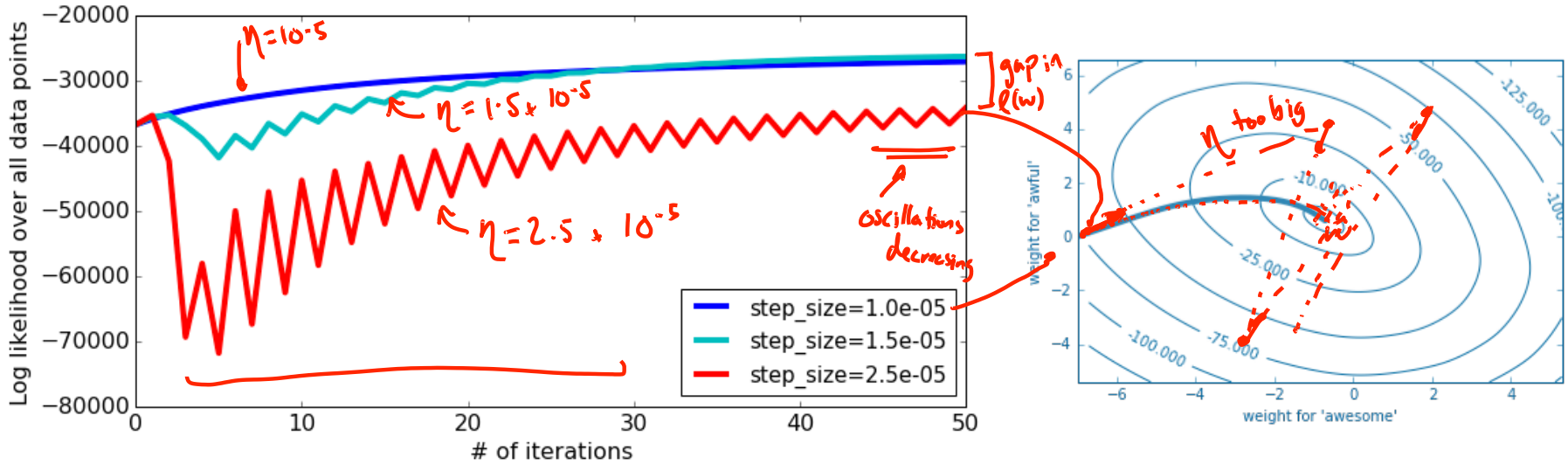
Careful with step sizes that are too large



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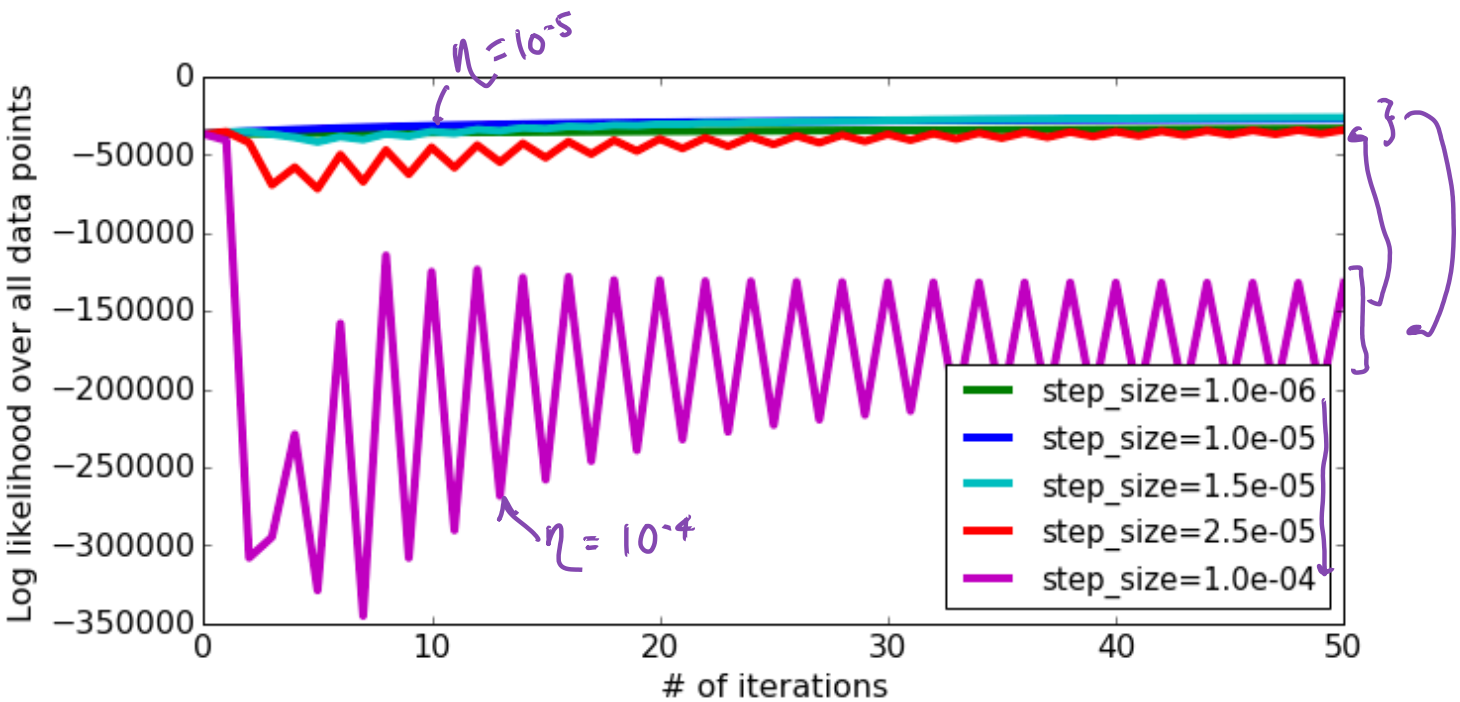
Very large step sizes can even cause divergence or wild oscillations



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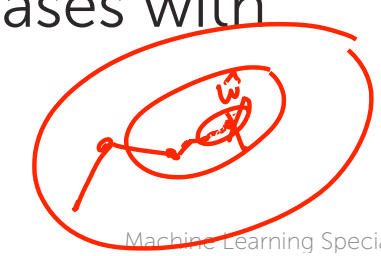
Simple rule of thumb for picking step size η

- Unfortunately, picking step size requires a lot of trial and error ☹
- Try a several values, exponentially spaced
  - **Goal**: plot learning curves to
    - find one η that is too small (smooth but moving too slowly)
    - find one η that is too large (oscillation or divergence)
- Try values in between to find "best" η

  ↳ exponentially space, pick one that leads best training data likelihood

- *Advanced tip*: can also try step size that decreases with iterations, e.g.,

$$\eta_t = \frac{\eta_0}{t}$$



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Deriving the gradient for logistic regression

VERY OPTIONAL

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Log-likelihood function

- Goal: choose coefficients **w** maximizing likelihood:

$$\ell(\mathbf{w}) = \prod_{i=1}^{N} P(y_i \mid \mathbf{x}_i, \mathbf{w})$$

- Math simplified by using log-likelihood – taking (natural) log:

$$\ell\ell(\mathbf{w}) = \ln \prod_{i=1}^{N} P(y_i \mid \mathbf{x}_i, \mathbf{w})$$

natural log

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The log trick, often used in ML…

- Products become sums:

  $\ln a \cdot b = \ln a + \ln b$ $\quad$ $\ln \frac{a}{b} = \ln a - \ln b$

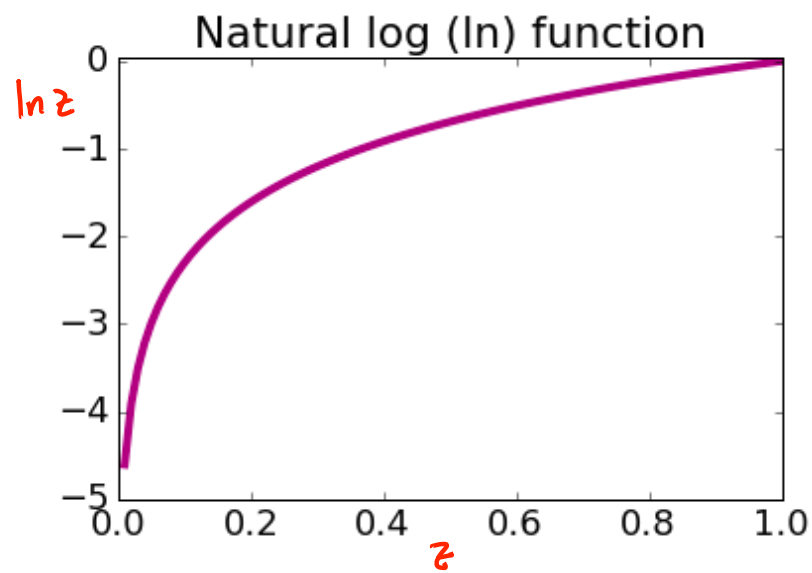
- Doesn't change maximum!
  - If $\hat{\mathbf{w}}$ maximizes f($\mathbf{w}$):

    $\hat{w} = \underbrace{\arg\max_{w}}_{\text{the } w \text{ that makes } f(w) \text{ largest}} f(w)$

  - Then $\hat{\mathbf{w}}_{\ln}$ maximizes ln(f($\mathbf{w}$)):

    $\hat{w}_{\ln} = \arg\max_{w} \ln(f(w))$

    $\hat{w} = \hat{w}_{\ln}$



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Insert next title slide before Slide 52, around 4:55 in PL7_DerivingtheGradient_1stEdit

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Expressing the log-likelihood

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Using log to turn products into sums

$$\ln \prod_{i=1}^{N} f_i = \sum_{i=1}^{N} \ln f_i$$

- The log of the product of likelihoods becomes the sum of the logs:

$$\ell\ell(\mathbf{w}) = \ln \prod_{i=1}^{N} P(y_i \mid \mathbf{x}_i, \mathbf{w})$$

$$= \sum_{i=1}^{N} \ln P(y_i \mid x_i, w)$$

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Rewriting log-likelihood

- For simpler math, we'll rewrite likelihood with indicators:

$$\ell\ell(\mathbf{w}) = \sum_{i=1}^{N} \ln P(y_i \mid \mathbf{x}_i, \mathbf{w})$$

$$= \sum_{i=1}^{N} \left[\mathbb{1}[y_i = +1] \ln P(y = +1 \mid \mathbf{x}_i, \mathbf{w}) + \mathbb{1}[y_i = -1] \ln P(y = -1 \mid \mathbf{x}_i, \mathbf{w}) \right]$$

if $y_i = +1$ ✓ 0

if $y_i = -1$ 0 ✓

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Insert next title slide before Slide 54, around 7:33 in PL7_DerivingtheGradient_1stEdit

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Deriving probability that y=-1 given x

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Logistic regression model: P(y=-1|x,w)

- Probability model predicts y=+1:

$$P(y=+1|\mathbf{x},\mathbf{w}) = \frac{1}{1+e^{-\mathbf{w}^T h(\mathbf{x})}}$$

- Probability model predicts y=-1:

$$P(y=-1|x,w) = 1 - P(y=+1|x,w) = 1 - \frac{1}{1+e^{-w^T h(x)}}$$

$$= \frac{1+e^{-w^T h(x)} - 1}{1+e^{-w^T h(x)}} = \frac{e^{-w^T h(x)}}{1+e^{-w^T h(x)}}$$

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Rewriting the log-likelihood

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Plugging in logistic function for 1 data point

$$P(y=+1 \mid \mathbf{x},\mathbf{w}) = \frac{1}{1+e^{-\mathbf{w}^\top h(\mathbf{x})}} \qquad P(y=-1 \mid \mathbf{x},\mathbf{w}) = \frac{e^{-\mathbf{w}^\top h(\mathbf{x})}}{1+e^{-\mathbf{w}^\top h(\mathbf{x})}}$$

$$\ell\ell(\mathbf{w}) = \mathbb{1}[y_i=+1]\ln P(y=+1 \mid \mathbf{x}_i,\mathbf{w}) + \mathbb{1}[y_i=-1]\ln P(y=-1 \mid \mathbf{x}_i,\mathbf{w})$$

$$= \mathbb{1}[y_i=+1]\ln\frac{1}{1+e^{-w^T h(x_i)}} + (1-\mathbb{1}[y_i=+1])\ln\frac{e^{-w^T h(x_i)}}{1+e^{-w^T h(x_i)}}$$

$$= -\mathbb{1}[y_i=+1]\ln(1+e^{-w^T h(x_i)}) + (1-\mathbb{1}[y_i=+1])\left[-w^T h(x_i) - \ln(1+e^{-w^T h(x_i)})\right]$$

$$= \underbrace{-(1-\mathbb{1}[y_i=+1])\,w^T h(x_i) - \ln(1+e^{-w^T h(x_i)})}_{\text{Simpler form}}$$

$$\ln e^a = a$$

$$\mathbb{1}[y_i=-1] = 1-\mathbb{1}[y_i=+1]$$

$$\ln\frac{1}{1+e^{-w^T h(x_i)}} = -\ln(1+e^{-w^T h(x_i)})$$

$$\ln\frac{e^{-w^T h(x_i)}}{1+e^{-w^T h(x_i)}} = \underbrace{\ln e^{-w^T h(x_i)}}_{-w^T h(x_i)} - \ln(1+e^{-w^T h(x_i)})$$

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Insert next title slide before Slide 56, around 16:56 in PL7_DerivingtheGradient_1stEdit

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Deriving gradient of log-likelihood

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Gradient for 1 data point

$$\ell\ell(\mathbf{w}) = -(1 - \mathbb{1}[y_i = +1])\mathbf{w}^\top h(\mathbf{x}_i) - \ln\left(1 + e^{-\mathbf{w}^\top h(\mathbf{x}_i)}\right)$$

$$\frac{\partial \ell\ell}{\partial w_j} = -(1 - \mathbb{1}[y_i = +1]) \frac{\partial}{\partial w_j} w^T h(x_i) - \frac{\partial}{\partial w_j} \ln\left(1 + e^{-w^T h(x_i)}\right)$$

$$= -(1 - \mathbb{1}[y_i = +1]) h_j(x_i) + h_j(x_i) P(y = -1 \mid x_i, w)$$

$$= h_j(x_i)\left[\mathbb{1}[y_i = +1] - P(y = +1 \mid x_i, w)\right]$$

$$\frac{\partial}{\partial w_j} w^T h(x_i) = h_j(x_i)$$

$$\frac{\partial}{\partial w_j} \ln\left(1 + e^{-w^T h(x_i)}\right)$$

$$= -h_j(x_i) \underbrace{\frac{e^{-w^T h(x_i)}}{1 + e^{-w^T h(x_i)}}}_{P(y=-1 \mid x_i, w)}$$

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Finally, gradient for all data points

- Gradient for one data point:

$$h_j(\mathbf{x}_i)\Big(\mathbb{1}[y_i = +1] - P(y = +1 \mid \mathbf{x}_i, \mathbf{w})\Big)$$

- Adding over data points:

$$\frac{\partial \ell\ell}{\partial w_j} = \sum_{i=1}^{N} h_j(x_i)\Big(\mathbb{1}[y_i = +1] - P(y = +1 \mid x_i, w)\Big)$$

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Summary of logistic regression classifier

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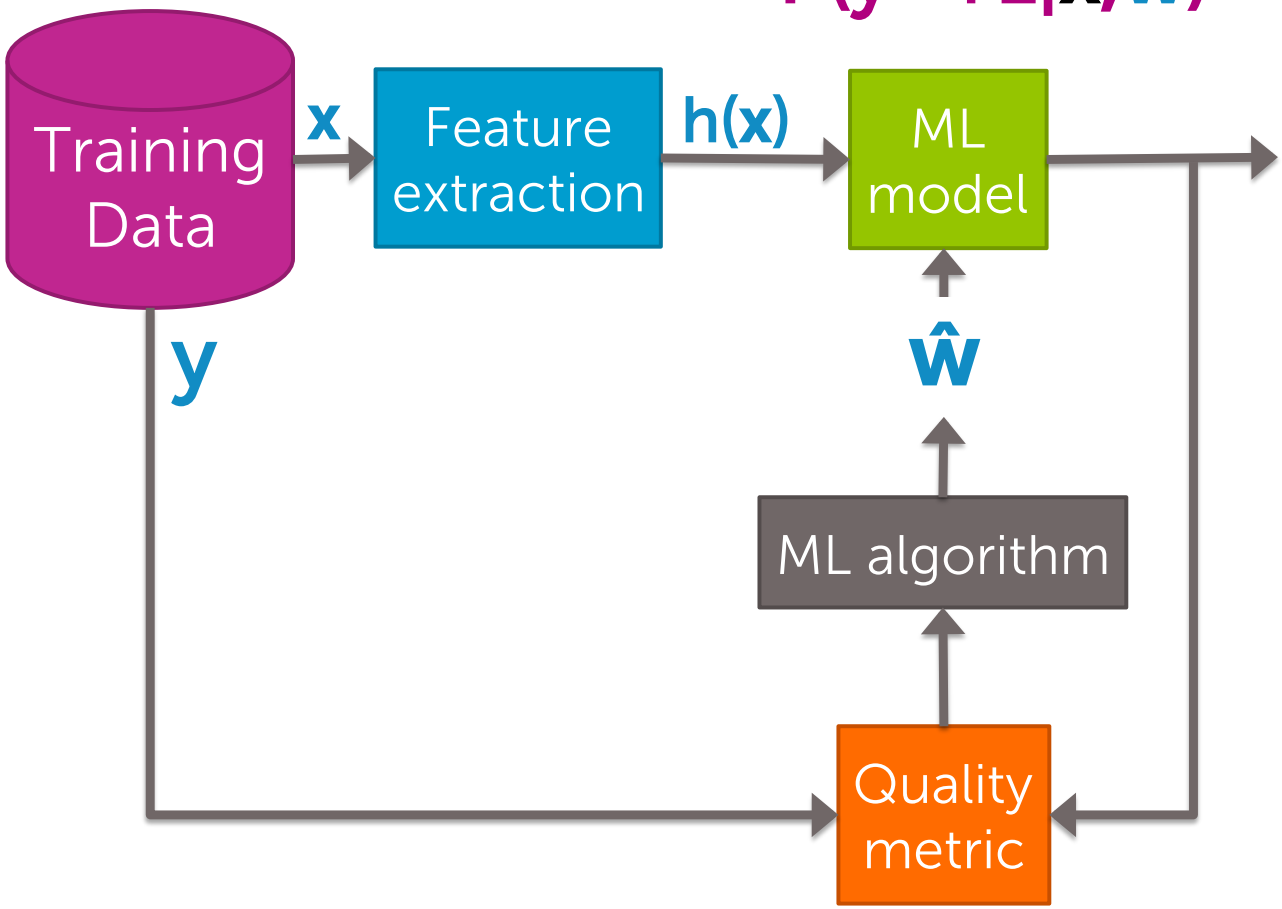
$$\hat{P}(y=+1|\mathbf{x},\hat{\mathbf{w}}) = \frac{1}{1+e^{-\hat{\mathbf{w}}^T h(\mathbf{x})}}$$



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What you can do now…

- Measure quality of a classifier using the likelihood function
- Interpret the likelihood function as the probability of the observed data
- Learn a logistic regression model with gradient descent
- (Optional) Derive the gradient descent update rule for logistic regression

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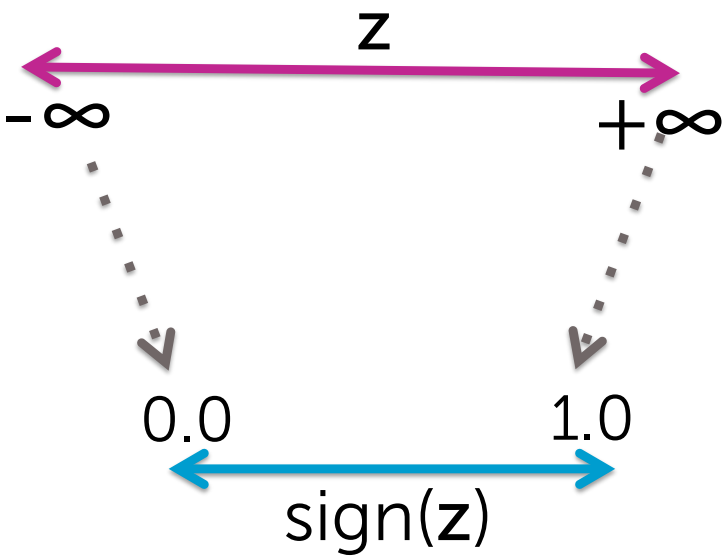
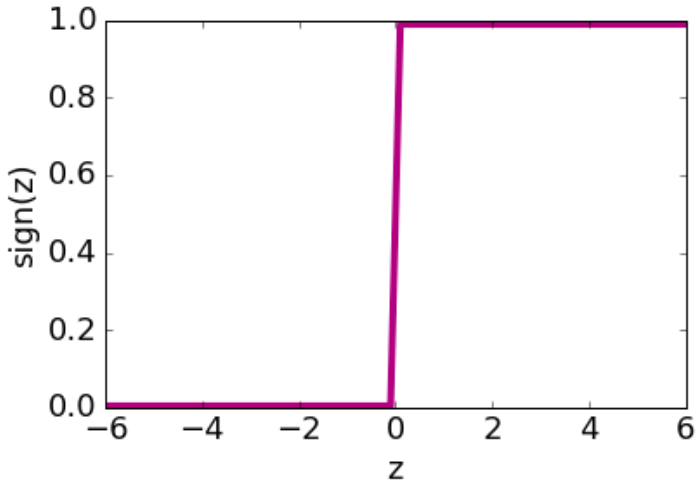
Simplest link function: sign(z)





$$sign(z) = \begin{cases} +1 & \text{if } z \geq 0 \\ -1 & \text{otherwise} \end{cases}$$

But, sign(z) only outputs -1 or +1, no probabilities in between

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Finding best coefficients

x[1] = #awesome	x[2] = #awful	y = sentiment
0	2	-1
3	3	-1
2	4	-1
0	3	-1
0	1	-1

x[1] = #awesome	x[2] = #awful	y = sentiment
2	1	+1
4	1	+1
1	1	+1
2	1	+1

$0.0 \leftarrow P(y=+1|\mathbf{x}_i,\hat{\mathbf{w}}) \rightarrow 1.0$

$\text{Score}(\mathbf{x}_i) = \hat{\mathbf{w}}^T h(\mathbf{x}_i)$

$-\infty \longleftrightarrow +\infty$

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Quality metric: probability of data

$$\hat{P}(y=+1|\mathbf{x},\hat{\mathbf{w}}) = \frac{1}{1+e^{-\hat{\mathbf{w}}^T h(\mathbf{x})}}$$

x[1] = #awesome	x[2] = #awful	y = sentiment
2	1	+1

If model good, should predict

Increase probability y=+1 when

Choose **w** to make

x[1] = #awesome	x[2] = #awful	y = sentiment
0	2	-1

If model good, should predict

Increase probability y=-1 when

Choose **w** to make

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Maximizing likelihood (probability of data)

Data point	x[1]	x[2]	y	Choose **w** to maximize
$\mathbf{x}_1, y_1$	2	1	+1	
$\mathbf{x}_2, y_2$	0	2	-1	
$\mathbf{x}_3, y_3$	3	3	-1	
$\mathbf{x}_4, y_4$	4	1	+1	
$\mathbf{x}_5, y_5$	1	1	+1	
$\mathbf{x}_6, y_6$	2	4	-1	
$\mathbf{x}_7, y_7$	0	3	-1	
$\mathbf{x}_8, y_8$	0	1	-1	
$\mathbf{x}_9, y_9$	2	1	+1	

Must combine into single measure of quality

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Learn logistic regression model with maximum likelihood estimation (MLE)

- Choose coefficients **w** that maximize likelihood:

$$\prod_{i=1}^{N} P(y_i \mid \mathbf{x}_i, \mathbf{w})$$

- No closed-form solution ➔ use gradient ascent

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